THE REPRESENTATION OF POSITIVE INTEGERS BY POSITIVE, DEFINITE QUADRATIC FORMS

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Abstract. In this paper, we survey approaches to the problem of the representation of a positive integer by an integral, positive, definite quadratic form. Naturally, this begins by looking at motivations of the p-adic completion of the rational numbers, leading us to the famous *Hensel's Lemma* and *Hasse-Minkowski Theorem*. Although these provide avenues for approaching our problem, obstructions in $\mathbb{Z}$ push us to consider *Siegel's Mass Formula* and an analytical approach for quadratic forms of $n \geq 5$ variables.

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1. Introduction

Quadratic forms are a significant object of study in number theory, having thousands of years of rich history, while remaining a lively area of research even today. A positive, definite, integral quadratic form[1], Q, in n variables is a polynomial of form:

$$Q(\vec{x}) = \sum_{1 \leq i \leq j \leq n} a_{ij} x_i x_j.$$

where $a_{ij} \in \mathbb{Z}$. We seek to understand whether Q is able to represent a positive integer m with $\vec{x} \in \mathbb{Z}^n$. That is, we want to see if there exists a choice of $\vec{x}$ such that $Q(\vec{x}) = m$. This question will form the basis for much of this paper, but it is certainly not the only question that arises in the study of such quadratic forms.

As one may notice, this first question is merely about the existence of a representation for a specific integer. So, what about the number of representations of a given integer? Though this is indeed a very interesting question, we will only briefly

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[1]For the remainder of this paper, when we talk about quadratic forms, we will specifically refer to positive, definite, integral quadratic forms, unless stated otherwise.

touch upon it. For the time being, denote the number of ways in which a quadratic form, Q, can represent an integer, m, by $r_Q(m) = |\{\vec{x} \in \mathbb{Z}^n : Q(\vec{x}) = m\}|$.

A simple example of a quadratic form is $Q(x_1, x_2) = x_1^2 + x_2^2$. Suppose we ask whether Q can represent 5. Here, we can guess and check until we see that for $x_1 = 1$ and $x_2 = 2$, $Q(1, 2) = 1^2 + 2^2 = 5$. Thus, the answer is yes. We can also see that $r_Q(m) = 8$.

However, can the quadratic form $Q(x_1, x_2, x_3) = 5x_1^2 + 8x_2^2 + 11x_3^2$ represent 512? The answer is also yes. Take $x_1 = 9, x_2 = 1$, and $x_3 = 3$. Yet, the path to solving this is not obvious at first glance. One may be inclined to try "guess and check" as used prior, but the number of choices for x_1, x_2, and x_3 increases considerably. Even more so, how can we be sure when a quadratic form cannot represent an integer without having to check all possible combinations? Or how will we calculate $r_Q(m)$? These questions suggest that a more systematic framework is needed to determine the representation of an integer by quadratic form. For this, we rely on the p-adic numbers.

2. p-adic Numbers

The p-adic numbers were first introduced by Kurt Hensel in 1897 in the study of modular arithmetic [8]. In what is to follow, we come to understand the p-adic numbers analytically. To view the algebraic construction of the p-adic numbers, see [8]. Where naturally we define the usual distance between two points in $\mathbb{R}$ by the Euclidean metric, the p-adic numbers are constructed on a completion of the rational numbers by a different metric, the p-adic metric. To begin building towards the definition of the p-adic metric, first consider an arbitrary prime, say p.

Definition 2.1 (p-adic Absolute Value)**.** Suppose $q \in \mathbb{Q}$ and $q \neq 0$, then

$$q = p^m \frac{a}{b}$$

where $m \in \mathbb{Z}$, $p \nmid a$, and $p \nmid b$. We define the p-adic absolute value of q to be

$$|q|_p = p^{-m}.$$

For $q = 0$, we define $|0|_p = p^{-\infty} = 0$.

Example 2.2. $|6|_3 = |3^1 \cdot 2|_3 = \frac{1}{3}$

Example 2.3. $|\frac{196}{13}|_7 = |7^2 \cdot \frac{4}{13}|_7 = \frac{1}{49}$

Example 2.4. $|\frac{3}{25}|_5 = |5^{-2} \cdot 3|_5 = 25$

Definition 2.5 (p-adic Metric)**.** The p-adic absolute value can be extended to a metric, d_p, for $\mathbb{Q}$. Suppose q_1, $q_2 \in \mathbb{Q}$, then

$$d_p(q_1, q_2) = |q_1 - q_2|_p$$

Theorem 2.6. *d_p satisfies the definition of a metric.*

Proof. Suppose $x, y, z \in \mathbb{Q}$.

(1) $d_p(x, y) \geq 0$: This is true, provided for any prime, p, $p^{-m} > 0$.

(2) $d_p(x, y) = 0 \iff x = y$: Suppose $d_p(x, y) = 0$. This is only true provided $|x - y|_p = 0$. That is, $x - y = 0$, so $x = y$.

(3) $d_p(x, y) = d_p(y, x)$: Suppose $x - y = w$ with $w \in \mathbb{Q}$. Then $y - x = -w$. Suppose w has factorization $w = p^m \frac{a}{b}$ with $p \nmid a$, and $p \nmid b$. Then $-w$ has factorization $w = p^m \frac{-a}{b}$ with $p \nmid -a$, and $p \nmid b$. Hence, $d_p(x, y) = d_p(y, x)$.

(4) $d_p(x, z) \leq d_p(x, y) + d_p(y, z)$: Take $x, y, z \in \mathbb{Q}$. Suppose $a = x - y$ and $b = y - z$. Suppose $d_p(x, y) = |a|_p$ and $d_p(y, z) = |b|_p$. Then it follows:

$$\begin{aligned} d_p(x, z) &= |x - z|_p \\ &= |a + b|_p \\ &\leq \max\{|a|_p, |b|_p\} \\ &= \max\{d_p(x, y), d_p(y, z)\}. \end{aligned}$$

This makes d_p what is called an *ultrametric*.

□

Example 2.7. $|100 - 2|_7 = |98|_7 = |7^2 \frac{2}{1}|_7 = \frac{1}{49}$

Example 2.8. $|\frac{3}{7} - \frac{1}{14}|_7 = |\frac{5}{14}|_7 = |7^{-1} \frac{5}{2}|_7 = 7$

Example 2.9. $|42069 - 37580|_{67} = |4489|_{67} = |67^2|_{67} = \frac{1}{67^2}$

With these three examples, we see that our intuition for what defines "closeness" with the Euclidean metric does not carry over to the p-adic metric. This imposes a very peculiar geometry and topology over the p-adic numbers, which can be read about here [11]. Interestingly, however niche as this metric may seem, its importance is shown with the following theorem.

Theorem 2.10 (Ostrowski's Theorem)**.** *Any non-trivial metric on $\mathbb{Q}$ is equivalent to one of the metrics $|\cdot|_p$ for prime p or $p = \infty$. See the proof in* [12].

When $p = \infty$, we have the classic Euclidean metric on $\mathbb{Q}$. Though this result is quite intriguing, we do not do much more with it in this paper, other than infer that this result suggests that we should pay attention to $|\cdot|_p$. Heeding this advice, we may consider what happens if we choose to complete $\mathbb{Q}$ with the p-adic metric.

The completion, $\mathbb{R}$ (of $\mathbb{Q}$), is the set of all equivalence classes of Cauchy sequences[2] of rational numbers under the Euclidean metric. Similarly, we can define the completion of $\mathbb{Q}$ with respect to $|\cdot|_p$ as the set of all equivalence classes of Cauchy sequences of rational numbers, which we call $\mathbb{Q}_p$. In this paper, we will not show this construction formally, so do see [8, 11]. By construction, we define $\mathbb{Q}_\infty$ to be $\mathbb{R}$.

What we will come to learn is that $\mathbb{Q}_p$ plays a vital role in our overlying question of the representation of an integer by a quadratic form. It also makes sense to acknowledge $\mathbb{Z}_p$, which is the completion of $\mathbb{Z}$ with respect to $|\cdot|_p$. Notably,

[2]A sequence (x_n) in a metric space (X, d) is a Cauchy sequence if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for any $m, n \geq N$, $d(x_m, x_n) < \varepsilon$.

$\mathbb{Z}_p \subset \mathbb{Q}_p$ where $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\}$. The interplay between $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{Q}_p$, and $\mathbb{Z}_p$ is crucially important. However, before we get to this discussion, we mention two important theorems to round off our talk about the p-adic numbers specifically.

Theorem 2.11 (Power Series Representation of a p-adic Integer)**.** *Suppose x is a p-adic integer. Then*

$$x = \sum_{i=0}^{\infty} x_i p^i$$

where $0 \leq x_i \leq p-1$.

Example 2.12. In $\mathbb{Q}_2$, $1 + 2 + 2^2 + ... = \frac{1}{1-2} = -1$.[3]

Theorem 2.13 (Power Series Representation of a p-adic Rational)**.** *Suppose x is a p-adic rational. Then*

$$x = \sum_{i=-m}^{\infty} x_i p^i$$

where $0 \leq x_i \leq p-1$ and $|x|_p = p^m$.

Example 2.14. In $\mathbb{Q}_5$, $\frac{1}{3} = 2 + 3 \cdot 5 + 1 \cdot 5^2 + 3 \cdot 5^3 + ...$

3. Hensel's Lemma and the Local-Global Principle

It is not immediately obvious how the construction of $\mathbb{Q}_p$ nor $\mathbb{Z}_p$ helps us with our general problem with quadratic forms. To see this, in this section, we will define and use the *Hasse-Minkowski Theorem* (i.e. the *Local-Global principle*).

Shortly after Hensel introduced the p-adic numbers, he proved a result that is crucial for the *Local-Global principle*. This result has become known as *Hensel's Lemma*, which we phrase with regards to $\mathbb{Z}_p$.

Lemma 3.1 (A Basic Form of Hensel's Lemma)**.** *Suppose $f(x)$ is a polynomial in $\mathbb{Z}_p[x]$ and for $x_0 \in \mathbb{Z}_p$, we have*

$$f(x_0) \equiv 0 \pmod{p}, f'(x_0) \not\equiv 0 \pmod{p}$$

then there exists a unique $y \in \mathbb{Z}_p$ such that $f(y) = 0$ in $\mathbb{Z}_p$ and $y \equiv x_0 \pmod{p}$. See the proof of this lemma in [**?**].

What *Hensel's Lemma* tells us is when we can effectively "lift" the roots of a polynomial $\pmod{p}$ to roots of the polynomial in $\mathbb{Z}_p$. It does this by iteratively lifting a solution $\pmod{p^k}$ to a solution $\pmod{p^{k+1}}$, constructing a sequence of roots that converges in $\mathbb{Z}_p$.

Example 3.2. Consider on $\mathbb{Z}_5$, $f(x) = x^2 + x + 5$. $f(4) = 4^2 + 4 + 5 = 25 \equiv 0 \pmod{5}$. However, $f'(4) = 2 \cdot 4 + 1 = 9 \not\equiv 0 \pmod{5}$. So, $x = 4$ lifts to a unique root $y \in \mathbb{Z}_5$ such that $f(y) = 0$ and $y \equiv 4 \pmod{5}$.

Though this basic form of *Hensel's Lemma* may work for polynomials of a single variable, most quadratic forms we care about will be multi-variable polynomials. Luckily, there is a direct analog of *Hensel's Lemma* for that.

[3] Moreover, in $\mathbb{Q}_p$, $\sum_{i=0}^{\infty} p^i = \frac{1}{1-p}$. See [2].

Lemma 3.3 (Hensel's Lemma for Multi-Variable Polynomials)**.** *Let $f(\vec{x})$ be a polynomial of n variables with p-adic integer coefficients. Let $x_1, \ldots, x_n$ be p-adic integers such that for some i with $1 \leq i \leq n$*

$$f(x_1, \ldots, x_n) \equiv 0 \pmod{p}, \qquad \frac{\partial f}{\partial x_i}(x_1, \ldots, x_n) \not\equiv 0 \pmod{p},$$

then there exist p-adic integers $y_1, \ldots, y_n$ such that

$$f(y_1, \ldots, y_n) = 0$$

and

$$y_i \equiv x_i \pmod{p} \quad \text{for } i = 1, \ldots, n.$$

The proof is detailed in [6].

In this case, suppose we care whether we can represent an integer m over $\mathbb{Z}_p$ by a quadratic form Q of n variables. Then by setting $f(\vec{x}) = Q(\vec{x}) - m$, this form of *Hensel's Lemma* allows us to determine when a solution of $f(\vec{x}) \equiv 0 \pmod{p}$ can be lifted to a solution over $\mathbb{Z}_p$.

Example 3.4. Take $Q(x, y) := 6x_1^2 + 7x_2^2$. Q can represent 12 over $\mathbb{Z}_5$.

Proof. First, define $f(x_1, x_2) := 6x_1^2 + 7x_2^2 - 12$. We see that for $x_1 = 3$ and $x_2 = 2$, $f(3, 2) = 6 \cdot 3^2 + 7 \cdot 2^2 - 12 = 70$. And, $70 \equiv 0 \pmod{5}$.
Next, we must confirm that either $\frac{\partial f}{\partial x_1} \not\equiv 0 \pmod{5}$ or $\frac{\partial f}{\partial x_2} \not\equiv 0 \pmod{5}$. Conveniently,

$$\frac{\partial f}{\partial x_1}(3, 2) = 36 \not\equiv 0 \pmod{5}$$

$$\frac{\partial f}{\partial x_2}(3, 2) = 28 \not\equiv 0 \pmod{5}$$

Hence, by *Hensel's Lemma*, there exists a $(y_1, y_2) \in \mathbb{Z}_5^2$ such that $f(y_1, y_2) = 0$ and $y_1 \equiv 3 \pmod{5}$ and $y_2 \equiv 2 \pmod{5}$. That is, there exists a $(y_1, y_2) \in \mathbb{Z}_5^2$ such that $Q(y_1, y_2) = 12$.

□

Example 3.5. This example will demonstrate a case when *Hensel's Lemma* would not guarantee lifting a solution. Consider the same quadratic form, $Q(x, y) := 6x_1^2 + 7x_2^2$. Suppose you are trying a determine whether Q can represent 66 over $\mathbb{Z}_3$. You would define $f(x, y) := 6x_1^2 + 7x_2^2 - 66$. A choice of $x_1 = 1$ and $x_2 = 3$ yields that $f(1, 3) = 3 \equiv 0 \pmod{3}$.
However,

$$\frac{\partial f}{\partial x_1}(1, 3) = 12 \equiv 0 \pmod{3}$$

$$\frac{\partial f}{\partial x_2}(1, 3) = 42 \equiv 0 \pmod{3}$$

.
So, *Hensel's Lemma* does not apply and does not imply that there is a solution over $\mathbb{Z}_3$. Yet, this does not mean that there is not a solution that could be lifted.

With this understood, we are now ready to approach the *Hasse-Minkowski Theorem*. *Hensel's Lemma* seems to be extremely useful in confirming representations over $\mathbb{Z}_p$, but what about $\mathbb{Z}$? Canonically, we refer to each $\mathbb{Z}_p$ as being local and $\mathbb{Z}$ as being global (with the same idea holding for $\mathbb{Q}_p$ and $\mathbb{Q}$). Connecting the existence of representations of an integer m by a quadratic form in $\mathbb{Z}_p$ to those in $\mathbb{Z}$ would be ideal to resolve our representation problem. This is exactly what the *Hasse-Minkowski Theorem* seeks to do, though with some caveats.

Proved originally by Hermann Minkowski, any quadratic form Q which represents 0 non-trivially over $\mathbb{R}$, represents 0 over $\mathbb{Q}$. What is actually being said here is that if Q represents 0 non-trivially over a completion of $\mathbb{Q}$ (in this case $\mathbb{R}$), then Q represents 0 over $\mathbb{Q}$. This was later generalized by Helmut Hasse, giving us the *Hasse-Minkowski Theorem*, which has become known as the *Local-Global principle*.

Theorem 3.6 (Hasse-Minkowski Theorem)**.** *Suppose $Q(\vec{x})$ is a quadratic form of n variables with rational coefficients, then $Q(\vec{x})$ represents $m \in \mathbb{Q}$ where $m \neq 0$ if and only if $Q(\vec{x})$ represents m in every completion of $\mathbb{Q}$. Namely, $\mathbb{R}$ and every $\mathbb{Q}_p$.*

Hasse was actually a student of Hensel, so it is no surprise to see the importance of *Hensel's Lemma* to Hasse and his theorem. We also see why this often referred to as the *Local-Global principle*. If we can ensure that Q represents m in every local $\mathbb{Q}_p$ and $\mathbb{R}$ (often by using *Hensel's Lemma*), then we know Q can also represent m in the global $\mathbb{Q}$.

Further, if our quadratic form can be written in diagonal form (which is possible by a linear change of variables for all quadratic forms with rational coefficients, not necessarily integer), $Q(\vec{x}) = \sum_{i=1}^{n} a_i x_i^2$ with $n \geq 2$, then representation in $\mathbb{Q}_p$ is guaranteed unless $p = 2$, $p|m$, or $p|a_i$ for some a_i. This is described in [4]. We extend this to $\mathbb{Z}_p$, so that our examples are expedited in the future.

Theorem 3.7 (Representation of an Integer in $\mathbb{Z}_p$)**.** *The representation of a nonzero integer m by quadratic form $Q(\vec{x})$ in $\mathbb{Z}_p$ is guaranteed when $Q(\vec{x})$ is in diagonal form, $p \neq 2$, $p \nmid a_i$ for all a_i, and $p \nmid m$.*

Proof. We show the case where $Q(\vec{x})$ is in diagonal form, $m \in \mathbb{Z}$, and we are looking for representation in $\mathbb{Z}_p$. However, the argument is exactly the same for rational numbers and $\mathbb{Q}_p$. Let $m \in \mathbb{Z}$. Suppose we have the quadratic form $Q(\vec{x})$ of n variables in diagonal form; we want to show there exists $\vec{x} \in \mathbb{Z}_p^n$ such that

$$Q(\vec{x}) = \sum_{i=1}^{n} a_i x_i^2 = m$$

for when $p \neq 2$, $p \nmid a_i$ for all a_i, and $p \nmid m$. Let us consider the first two terms of this sum,

$$a_1 x_1^2 + a_2 x_2^2$$

If we can show that there exists x_1, x_2 such that $a_1 x_1^2 + a_2 x_2^2 \equiv m \pmod{p}$, this will imply (by taking $x_3, x_4, ..., x_n = 0$) that there exists a $\vec{x} \in \mathbb{Z}_p^n$ such that $Q(\vec{x}) = \sum_{i=1}^{n} a_i x_i^2 = m$ by *Hensel's Lemma*.

Using the technique of [4], consider the sets $A := \{a_1 x_1^2 \pmod{p} : 0 \leq x_1 \leq p-1\}$ and $B := \{m - a_2 x_2^2 \pmod{p} : 0 \leq x_2 \leq p-1\}$. Letting $x_1, x_2 \in \{0, 1, ..., p-1\}$ effectively accounts for all possible choices of x_1, x_2. Then note for $x_1 \neq 0, x_1^2 \equiv$

$(-x_1)^2 \pmod p$. For example, $1^2 \equiv (p-1)^2 \pmod p$. Then this means, when including $x_1 = 0$, $|A| = 1 + \frac{p-1}{2} = \frac{p+1}{2}$. Similarly, $|A| = |B|$.

However, the total number of residue classes is p which is less than $|A| + |B| = p + 1$. So, $A \cap B \neq \emptyset$. Thus, by the pigeonhole principle, there must exist x_1, x_2 such that $a_1 x_1^2 \equiv m - a_2 x_2^2 \pmod p$. That is,

$$a_1 x_1^2 + a_2 x_2^2 \equiv m \pmod p$$

as needed. Also, provided $p \nmid m$, then at least one of x_1 or x_2 must be nonzero. Next, defining

$$f(\vec{x}) = \left(\sum_{i=1}^{n} a_i x_i^2 \right) - m$$

we see that for the choice of $\vec{x} = (x_1, x_2, 0, 0, ..., 0)$,

$$f(\vec{x}) \equiv 0 \pmod p$$

and

$$\frac{\partial f}{\partial x_1}(x_1, x_2) = 2a_1 x_1 \not\equiv 0 \pmod p$$

or

$$\frac{\partial f}{\partial x_2}(x_1, x_2) = 2a_2 x_2 \not\equiv 0 \pmod p$$

as $p \neq 2$ and $p \nmid a_i$ for all a_i. Hence, by *Hensel's Lemma* $\vec{x}$ lifts to a representation over $\mathbb{Z}_p$.

□

Thankfully, this makes checking whether a quadratic form represents an integer a doable task. Instead of looking at an infinite number of completions of $\mathbb{Q}$, we need only check a finite amount. However, do notice that the *Hasse-Minkowski Theorem* is specific to finding global solutions in $\mathbb{Q}$. Unfortunately, finding solutions in all $\mathbb{Z}_p$ and $\mathbb{R}$ does not suffice for a solution in $\mathbb{Z}$, but it does for $\mathbb{Q}$. The following examples look at this shortcoming.

Example 3.8. Can $x_1^2 + x_2^2 = 3$? This is the famous quadratic form studied by Fermat. In this example, we show that this quadratic form cannot represent 3. It is abundantly clear that there is no solution of $x_1, x_2 \in \mathbb{Z}$ such that $x_1^2 + x_2^2 = 3$, but it is also true that representation fails in $\mathbb{Z}_p$.

3 can be represented over $\mathbb{R}$, provided $0^2 + (\sqrt{3})^2 = 3$. Also from 3.7, solvability in all $\mathbb{Z}_p$ where $p \neq 2, 3$ is true.

Let us look at $\mathbb{Z}_2$. For the sake of contradiction, suppose 3 can be represented over $\mathbb{Z}_2$. Then we must have a solution (x_1, x_2), for $x_1^2 + x_2^2 \equiv 3 \pmod{2^n}$ for all n. However, for any sum of squares, $x_1^2 + x_2^2 \equiv 0 \pmod 4$ or $x_1^2 + x_2^2 \equiv 1 \pmod 4$ or $x_1^2 + x_2^2 \equiv 2 \pmod 4$. That is, $x_1^2 + x_2^2 \not\equiv 3 \pmod 4$. This is our contradiction as desired. Hence, there cannot be a solution in $\mathbb{Z}_2$ and no solution in $\mathbb{Z}$.

Example 3.9. Can $x_1^2 + 3x_2^2 + 5x_3^2 = 33$? Representation is guaranteed in all $\mathbb{Z}_p$ where $p \neq 2, 3, 5, 11$ by 3.7.

Notice that if we try to apply *Hensel's Lemma* for $\mathbb{Z}_2$, this will never work because the partial derivatives are always divisible by 2. To tackle this, we will follow a similar strategy as [4]. Let $x_2 = x_3 = 0$. Then we have that $x_1^2 = 33$. Hence, as $33 \equiv 1 \pmod 8$, then 33 is a square in $\mathbb{Z}_2$. So, $x_1^2 + 3 \cdot 0^2 + 5 \cdot 0^2 = 33$ is solvable in $\mathbb{Z}_2$, and there exists a solution in $\mathbb{Z}_2$.[4]

Next, we check $\mathbb{Z}_3$. Note, for $x_1 = 1, x_2 = 0$, and $x_3 = 2$,

$$1^2 + 3 \cdot 0^2 + 5 \cdot 2^2 - 33 = -12 \equiv 0 \pmod 3$$

and

$$2x_1 = 2 \not\equiv 0 \pmod 3$$

.

Thus, *Hensel's Lemma* applies, and there is a solution in $\mathbb{Z}_3$. Further, we check $\mathbb{Z}_5$. Note, for $x_1 = 0, x_2 = 1$, and $x_3 = 3$,

$$0^2 + 3 \cdot 1^2 + 5 \cdot 3^2 - 33 = 15 \equiv 0 \pmod 5$$

and

$$6x_2 = 6 \not\equiv 0 \pmod 5$$

So, *Hensel's Lemma* applies, and there is a solution in $\mathbb{Z}_5$. Finally, we check $\mathbb{Z}_{11}$. Note, for $x_1 = 1, x_2 = 2$, and $x_3 = 2$,

$$1^2 + 3 \cdot 2^2 + 5 \cdot 2^2 - 33 = 0 \equiv 0 \pmod{11}$$

and

$$6x_2 = 12 \not\equiv 0 \pmod{11}$$

Subtly, this last confirmation in $\mathbb{Z}_{11}$ gave us our global solution. However, to proceed with the example's intent, there is clearly a solution in $\mathbb{R}$, as $(\sqrt{10})^2 + 3 \cdot 1^2 + 5 \cdot 2^2 = 33$. Therefore, by the *Hasse-Minkowski Theorem*, as we have shown in each $\mathbb{Z}_p$ (realistically, $\mathbb{Q}_p$) that we can represent 33, then there exists a representation of 33 in $\mathbb{Q}$. In this case, we get lucky and it is also in $\mathbb{Z}$, as $1^2 + 3 \cdot 2^2 + 5 \cdot 2^2 = 33$ (just as seen with our choice for $\mathbb{Z}_{11}$).

Example 3.10. Can $x_1^2 + 19x_2^2 = 11$? Representation is guaranteed in all $\mathbb{Z}_p$ where $p \neq 2, 11, 19$ by 3.7.

Applying the same strategy as above, let $x_1 = 0$. Then we have that $19x_2^2 = 11$ or $x_2^2 = \frac{11}{19}$. Hence, as $11/19 \equiv 1 \pmod 8$, then $\frac{11}{19}$ is a square in $\mathbb{Z}_2$. So, $0^2 + 19x_2^2 = 11$ is solvable in $\mathbb{Z}_2$, and there exists a solution in $\mathbb{Z}_2$.

Next, we check $\mathbb{Z}_{11}$. Note, for $x_1 = 5$ and $x_2 = 1$,

$$5^2 + 19 \cdot 1^2 - 11 = 33 \equiv 0 \pmod{11}$$

and

$$2x_1 = 10 \not\equiv 0 \pmod{11}$$

.

Thus, *Hensel's Lemma* applies, and there is a solution in $\mathbb{Z}_{11}$. Further, we check $\mathbb{Z}_{19}$. For $x_1 = 7$ and $x_2 = 0$,

[4]This comes from a theorem that says that if $u \equiv 1 \pmod 8$, then u is a square in $\mathbb{Z}_2$

$$7^2 + 19 \cdot 0^2 - 11 = 38 \equiv 0 \pmod{19}$$

and

$$2x_1 = 14 \not\equiv 0 \pmod{19}$$

Therefore, *Hensel's Lemma* applies, and we can represent 11 in $\mathbb{Z}_{19}$. Also, there is a solution in $\mathbb{R}$, as $(\sqrt{11})^2 + 19 \cdot 0^2 = 11$. Therefore, by the *Hasse-Minkowski Theorem*, then there exists a representation of 11 in $\mathbb{Q}$. Unlike our last example however, there is no solution over $\mathbb{Z}$. Yet, there is a solution over $\mathbb{Q}$.

$$\left(\frac{5}{2}\right)^2 + 19 \cdot \left(\frac{1}{2}\right)^2 = \frac{44}{4} = 11$$

As we can see from these examples, representation over $\mathbb{Z}_p$ and $\mathbb{R}$ is not sufficient to ensure that a solution will exist with $\mathbb{Z}$. Though this strategy of employing the *Local-Global principle* works well, the rest of this paper will be dedicated to other techniques that allow us to determine whether a quadratic form represents an integer.

4. Siegel's Mass Formula

This trouble of not being able to fully guarantee the representation of an integer over $\mathbb{Z}$ is largely at the consequence of our attempt to try to connect our local to global. Though local examination of quadratic forms is beneficial, many issues arise in being able to distinguish different integral, quadratic forms from one another, leading to the complications we saw with the examples of last section.

Definition 4.1 (Equivalent Quadratic forms)**.** Take two quadratic forms Q_1 and Q_2 with n variables.

(1) Such quadratic forms are equivalent over $\mathbb{Z}$ if there exists $M \in \mathrm{GL}_n(\mathbb{Z})$[5] such that $Q_2(\vec{x}) = Q_1(M\vec{x})$ for all $\vec{x} \in \mathbb{Z}^n$. ($Q_1 \sim_{\mathbb{Z}} Q_2$)
(2) For a chosen prime p, such quadratic forms are equivalent over $\mathbb{Z}_p$ if there exists $M \in \mathrm{GL}_n(\mathbb{Z}_p)$ such that $Q_2(\vec{x}) = Q_1(M\vec{x})$ for all $\vec{x} \in \mathbb{Z}_p^n$. ($Q_1 \sim_{\mathbb{Z}_p} Q_2$)

We can also think of the ways in which a quadratic form is equivalent to itself.

Definition 4.2 (Automorphism Group of a Quadratic Form)**.** The *automorphism group* of Q, denoted $\mathrm{Aut}(Q)$, is the set

$$\mathrm{Aut}(Q) = \{\, M \in \mathrm{GL}_n(\mathbb{Z}) : Q(M\vec{x}) = Q(\vec{x}) \text{ for all } \vec{x} \in \mathbb{Z}^n \,\}.$$

When looking at the local equivalence between quadratic forms, we often will see that a quadratic form is locally equivalent to many other quadratic forms. We define the set of locally equivalent quadratic forms to be the *genus* of the form, denoted $Gen(Q)$.

Definition 4.3 (Genus of Quadratic Form)**.** The *genus* of Q_1, denoted $\mathrm{Gen}(Q_1)$, is the set of all integral quadratic forms Q_2 such that:

(1) $Q_2 \sim_{\mathbb{R}} Q_1$, and
(2) $Q_2 \sim_{\mathbb{Z}_p} Q_1$ for every prime p.

[5] $\mathrm{GL}_n(\mathbb{Z})$ is all $n \times n$ matrices with integer entries which are invertible over $\mathbb{Z}$.

In an ideal world, $Gen(Q)$ would give us enough information to understand whether an integer can be represented for a chosen $Q_0 \in Gen(Q)$. Unfortunately, this is not the case, and it is here where the idea of local and global equivalence plays a large role. Partitioning $Gen(Q)$ are what we call the *classes* of Q.

Definition 4.4 (Class of $Gen(Q)$)**.** The *class* of $Q_1 \in Gen(Q)$, denoted $[Q_1]$, is the set of all quadratic forms Q_0 such that $Q_0 \sim_{\mathbb{Z}} Q_1$.

Thus, the *class* of a quadratic form is the collection of all quadratic forms that are globally equivalent to a given form Q. As we can see, this is a stricter requirement than what is required to be in $Gen(Q)$. What this has systematically set up is the way in which we can classify all of the quadratic forms residing in a provided *genus*. The next theorem makes this task approachable.

Theorem 4.5 (Hermite's Theorem)**.** *Each* genus *contains only finitely many classes.*

Taking this definition and theorem into account for a $Gen(Q)$, we have that

$$Gen(Q) = \bigcup_{i=1}^{k} [Q_i]$$

where k is the finite number of *classes* in $Gen(Q)$. But now a natural question arises. If we can understand the *classes* that reside in $Gen(Q)$, does this provide us with avenue to make any claims about the global representations of the *genus* more broadly? What about the local representations too? These are the questions which lead us to what is called *Siegel's Mass Formula* for positive, definite quadratic forms.

Theorem 4.6 (Siegel's Mass Formula)**.** *Provided $Gen(Q)$ is a disjoint, finite union of classes represented by forms Q_i, then*

$$r_{\mathrm{Gen}(Q)}(m) := \frac{\displaystyle\sum_{Q_i} \frac{r_{Q_i}(m)}{|\mathrm{Aut}(Q_i)|}}{\displaystyle\sum_{Q_i} \frac{1}{|\mathrm{Aut}(Q_i)|}} = \prod_{p \leq \infty} \beta_p(m),$$

with $\beta_p(m)$ as the local representation densities at p. See a proof in [7].

$\beta_p(m)$ defines how many local representations m has in $\mathbb{Z}_p$. Thus, *Siegel's Mass Formula* provides a way to connect the number of local representations of a quadratic form to the weighted average number of global representations across the classes in $Gen(Q)$. It does this by weighting by the size of $\mathrm{Aut}(Q_i)$, as a quadratic form that is globally equivalent to several other quadratic forms will over-count the number of global representations. As a result, this formula helps us understand why local representation need not always imply global representation, as with the examples of last chapter. In practice, determining $|Aut(Q)|$ and $\beta_p(m)$ is not straightforward, but *Siegel's Mass Formula* provides an avenue to prove classic results as a consequence of the following corollary.

Corollary 4.7. *Suppose Q is a positive, definite quadratic form. If there is only 1* class *in $Gen(Q)$, then m is locally represented by Q if and if only m is globally represented by Q.*

Proof. Assume we have the quadratic form Q, and $Gen(Q)$ has only 1 class. Let us denote this class by $[Q]$ (provided Q must reside in this class as there is only 1 class for all of $Gen(Q)$).
By *Siegel's Mass Formula*,

$$\prod_{p\leq\infty}\beta_p(m) = \frac{\displaystyle\sum_{Q_i}\frac{r_{Q_i}(m)}{|\mathrm{Aut}(Q_i)|}}{\displaystyle\sum_{Q_i}\frac{1}{|\mathrm{Aut}(Q_i)|}}$$

$$= \frac{\dfrac{r_Q(m)}{|\mathrm{Aut}(Q)|}}{\dfrac{1}{|\mathrm{Aut}(Q)|}}$$

$$= r_Q(m).$$

That is, $\prod_{p\leq\infty}\beta_p(m) > 0$ if and only if $r_Q(m) > 0$. So, local and global representation imply one another.

□

Theorem 4.8 (Fermat's Sum of Squares Theorem)**.** *A prime number $m > 2$ can be represented by $x_1^2 + x_2^2$ if and only if $m \equiv 1 \pmod 4$.*

Proof. Suppose m is a prime number. The quadratic form that we care about, $x_1^2 + x_2^2$, has only one class in its genus. Thus, by *Siegel's Mass Formula* and 4.7, local representation will imply global representation. So, let us keep that in mind.

$\Rightarrow$ Assume m is globally represented by $x_1^2 + x_2^2$. Then by the *Hasse-Minkowski Theorem* and 4.7, m must be represented at every $\mathbb{Z}_p$ and $\mathbb{R}$. By our result of 3.7, local solvability is immediate for all $\mathbb{Z}_p$ when $p \neq 2$ and for $\mathbb{R}$, regardless of our choice of m. So, the only completion where there could be any obstruction is $\mathbb{Z}_2$. Let us look at when $p = 2$. Since $m > 2$ and prime, m is odd. Consequently, we must have that either $m \equiv 1 \pmod 4$ or $m \equiv 3 \pmod 4$. However, by the same argument as in examples 3.8 and 3.9, solvability in $\mathbb{Z}_2$ is only guaranteed when $m \equiv 1 \pmod 4$. Hence, $m \equiv 1 \pmod 4$, as desired.

$\Leftarrow$ Assume $m \equiv 1 \pmod 4$. Then as before, local representation will be guaranteed at all $\mathbb{Z}_p$ and $\mathbb{R}$. Thus by 4.7, this local representation at all $\mathbb{Z}_p$ implies global representation in $\mathbb{Z}$. That is, m can be represented globally by $x_1^2 + x_2^2$.

□

Theorem 4.9 (Lagrange's Four Squares Theorem)**.** *Any nonnegative integer m can be written as a sum of four integer squares.*

Proof. Suppose m is a nonnegative integer. The quadratic form for this problem is $x_1^2 + x_2^2 + x_3^2 + x_4^2$, which has only one class in its genus. By *Siegel's Mass Formula* and 4.7, local representation will imply global representation. We will try to confirm local representation in all $\mathbb{Z}_p$ and $\mathbb{R}$, so that this will effectively imply

global representation over $\mathbb{Z}$. Local representation is immediately guaranteed in $\mathbb{R}$ and all $\mathbb{Z}_p$ where $p \neq 2$. We will split this into two cases (though truly 8 cases) over $\mathbb{Z}_2$.

Case 1: $m \not\equiv 0 \pmod 8$

There are actually seven cases being dealt with here. Namely, we have when $m \equiv 1 \pmod 8$, $m \equiv 2 \pmod 8$, $\ldots m \equiv 7 \pmod 8$. Note that for any of these cases, we can make a choice of x_2, x_3, and x_4 such that $x_1^2 = m - (x_2^2 + x_3^2 + x_4^2) \equiv 1 \pmod 8$, as shown in the table below.

$m \pmod 8$	Choice of $(x_2^2, x_3^2, x_4^2) \pmod 8$
1	$(0,0,0)$
2	$(1,0,0)$
3	$(1,1,0)$
4	$(1,1,1)$
5	$(4,0,0)$
6	$(1,4,0)$
7	$(1,1,4)$

Then by the same argument as 3.8, x_1^2 is a square in $\mathbb{Z}_2$, leading us to see that $x_1^2 + x_2^2 + x_3^2 + x_4^2$ can represent m in $\mathbb{Z}_2$.

Case 2: $m \equiv 0 \pmod 8$

If $m \equiv 0 \pmod 8$, then $m = 4^k \cdot n$, where k is maximal and $n \not\equiv 0 \pmod 8$. Since we know $n \not\equiv 0 \pmod 8$, by **Case 1**, we can represent n in $\mathbb{Z}_2$. Multiplying the choices of x_1, x_2, x_3 and x_4 for n by 2^k gives us then a representation of m in $\mathbb{Z}_2$, as desired.

Hence, m is locally represented at all $\mathbb{Z}_p$ and $\mathbb{R}$. Therefore, by 4.7, global representation is guaranteed. That is, any nonnegative integer can be represented by the sum of four integer squares. □

5. An Analytical Approach for $n \geq 5$ variables

Finally, our last stop in this journey of learning the tools to exploit this local to global interplay of our problem takes us to a purely analytical approach: the *Theta function*. The original *Theta function*, seen below, was detailed in a letter from Euler to Goldbach.

$$\Theta(\tau) = \sum_{m=-\infty}^{\infty} e^{\pi i m^2 \tau}$$

At first glance, this seems to have nothing to do with quadratic forms. But do notice,

$$\begin{aligned}\Theta^2(\tau) &= \left(\sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau}\right)\left(\sum_{m=-\infty}^{\infty} e^{\pi i m^2 \tau}\right)\\ &= \sum_{n=-\infty}^{\infty}\sum_{m=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{\pi i m^2 \tau}\\ &= \sum_{n=-\infty}^{\infty}\sum_{m=-\infty}^{\infty} e^{\pi i (n^2+m^2)\tau}\\ &= \sum_{k=0}^{\infty} r_Q(k) e^{\pi i k \tau}\end{aligned}$$

where $Q(x_1, x_2) = x_1^2 + x_2^2$. So, the *Theta function* can effectively encode the number of representations of an integer by a given quadratic form, with this example being the classic quadratic form $x_1^2 + x_2^2$. Even more so, the *Theta function* can be used to reprove the results of 4.8 and 4.9, see [5, 13, 14]. However, more generally when we consider an arbitrary quadratic form, Q, we think of the *Theta function* as

$$\Theta_Q(\tau) = \sum_{m=0}^{\infty} r_Q(m) e^{2\pi i m \tau}$$

By being able to understand the coefficients $r_Q(m)$, we are not only able to understand the number of representations of an integer m, but also, whether that integer can be represented whatsoever. Namely, if we can be confident (that for a chosen m) the *Theta function* has coefficient $r_Q(m) > 0$, then Q must globally represent m. In this section, we will examine a methodology that ensures representation of m when m is sufficiently large and Q is in $n \geq 5$ variables.

As a result of its construction, $\Theta_Q(\tau)$ behaves as a holomorphic (complex-differentiable) function on the upper-half complex plane, $\mathbb{H}$. This then leads us to the definition of a modular form and a related statement about the *Theta function*.

Definition 5.1. Let $\Gamma_0(N)$[6] be the congruence subgroup of level N. Let χ be a Dirichlet character[7] modulo N. A holomorphic function $f : \mathbb{H} \to \mathbb{C}$ is called a *modular form of weight k, level N, and character χ* if the following hold:

(1) For all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N)$ and $\tau \in \mathbb{H}$, $f(\gamma\tau) = \chi(d)\,(c\tau + d)^k\, f(\tau)$.

(2) f is holomorphic at every cusp[8] of $\Gamma_0(N)$.

[6]Precisely, this is defined as $\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \,\middle|\, c \equiv 0 \pmod{N} \right\}$.

[7]See [10] for the precise definition of a Dirichlet character.

[8]A *cusp* of $\Gamma_0(N)$ is a point in the extended rational line $\mathbb{Q} \cup \{\infty\}$ considered up to the action of Γ by fractional linear transformations.

Theorem 5.2. *Suppose Q is a positive, definite, integral quadratic form of n variables. There exists $\Gamma_0(N) \subset SL_2(\mathbb{Z})$ such that*

$$\Theta_Q(\gamma\tau) = \chi_Q(d)\,(c\tau + d)^{n/2}\,\Theta_Q(\tau),$$

for some character χ_Q on $(\mathbb{Z}/N\mathbb{Z})^\times$.

Under this definition, we can recognize that the *Theta function* is a modular form of weight $\frac{n}{2}$ for $\Gamma_0(N)$. In this paper, we do not do the justice for which this realization deserves. For more information on this and modular forms, see [9] which provides sources that examine this more deeply.

But what does this mean for our problem? The *Theta function* fulfilling the definition of modular form actually provides us with a new way to understand the coefficients of $r_Q(m)$. This is done by recognizing $r_Q(m)$ as the sum of the coefficients of two series for which we can decompose the *Theta function* into. These are obtained from the Fourier Expansion of the *Eisenstein series* and *Cusp form*.

Definition 5.3. The *Eisenstein series of weight k* is given by

$$E_k(\tau) = \sum_{(m,n)\in\mathbb{Z}^2-\{(0,0)\}} \frac{1}{(m+n\tau)^k}, \qquad \tau \in \mathbb{H}.$$

Definition 5.4 (Cusp form)**.** A modular form $f(\tau)$ of weight k, level N, and character χ is a *cusp form* if f is holomorphic at every cusp of $\Gamma_0(N)$ and vanishes at all cusps.

Theorem 5.5 (Eisenstein and Cusp Form Decomposition)**.** *Every modular form f can be written uniquely in the form*

$$f(\tau) = E(\tau) + g(\tau),$$

where $E(\tau) \in E_k(\Gamma_0(N), \chi)$ and $g(\tau) \in S_k(\Gamma_0(N), \chi)$. See [1].

Therefore by 5.5, since the *Theta function* is a modular form, we can decompose it into an Eisenstein series and cusp form. That is, there exist $E(\tau) \in E_{n/2}(\Gamma_0(N), \chi)$ and $g(\tau) \in S_{n/2}(\Gamma_0(N), \chi)$ such that

$$\Theta_Q(\tau) = E(\tau) + g(\tau)$$

for which the Fourier expansion of each yields $r_Q(m) = a_E(m) + a(m)$. Consequently, if we can understand the coefficients of the Eisenstein series and Cusp form, then we undoubtedly understand $r_Q(m)$ of the original *Theta function*. However, like most things in life, this is easier said than done. One can computationally calculate the coefficients of such series, but from an analytical perspective, it makes sense to see if we can establish any bounds on these coefficients hailing from each respective series. This approach forms the basis of what is the *Tartakowsky-Kloosterman Theorem*.

Suppose we have a quadratic form of n variables, with $r_Q(m)$ given by an associated *Theta function*. For cusp forms of weight $\frac{n}{2}$, $|a(m)| << m^{\frac{n}{4}}$, with $a(m)$ possibly negative [1, 9]. Alternatively, for the Eisenstein series, either $a_E(m) = 0$ or $a_E(m) \approx m^{\frac{n}{2}-1}$ [1, 9]. Thus, $m^{\frac{n}{2}-1} > m^{\frac{n}{4}}$, precisely when $n \geq 5$. Hence, for

sufficiently large m and $a_E(m) > 0$, we must have that $r_Q(m) > 0$ [9]. Yet, then the question becomes: when can we guarantee that $a_E(m) \neq 0$?

Conveniently, Siegel (from 4.6) made a beautiful insight about $a_E(m)$:

$$a_E(m) = \prod_{p \leq \infty} \beta_p(m)$$

The Fourier coefficients of the Eisenstein series are precisely the product of the local representation densities from 4.6! So, if we are able to ensure that a quadratic form of $n \geq 5$ variables is able to represent a sufficiently large m over all $\mathbb{Z}_p$, then it must also globally represent m. This is exactly what the *Tartakowsky-Kloosterman Theorem* states.

Theorem 5.6 (Tartakowsky-Kloosterman Theorem)**.** *Suppose Q is a positive, definite, integral quadratic form in $n \geq 5$ variables. Then every sufficiently large m that is represented at all $\mathbb{Z}_p$ is represented globally over $\mathbb{Z}$.*

This result speaks volumes to the pervasiveness of the *Hasse-Minkowski Theorem* and serves as a nice way to conclude the material of this paper. Although the motivation for study of the *Theta function* originated from a purely analytical nature, it leads to this gorgeous result combining analytical approaches with the algebraic viewpoint of the *Local-Global principle*. A significant body of work today is dedicated at this intersection - and it is clear that this is for good reason.

6. Conclusion

As can be seen from this paper, the representation of an integer by an arbitrary quadratic form is most certainly not a simple question. At the heart of this problem is the interplay of the local p-adic $\mathbb{Z}_p$ and $\mathbb{Q}_p$, with the global $\mathbb{Z}$ and $\mathbb{Q}$. Even when local representation in $\mathbb{Q}_p$ nicely confirms representation in $\mathbb{Q}$, the same does not hold true for $\mathbb{Z}$.

As such, we often look for other methods when we want to understand this problem for integral quadratic forms, with some of these strategies arising in analytical methods. This problem is still a major area of research in number theory today, with important theorems such as the *290 Theorem* being proven recently. However, a generalizable methodology for understanding this question for an arbitrary integral quadratic form (even disregarding positive definiteness) is not settled. In this paper, we have come to understand only but a small subset of the mature research and history related to this problem, and even with the plethora of other techniques out there, quadratic forms remain a difficult (but important) challenge for us all.

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[9] There is actually much more to say about the bounds here. [1] provides a great explanation on this with some computational examples (though on quaternary forms).

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